



RippleWealth: AI-Driven Investment Risk Intelligence Platform

Sprint 1 Deliverable:

Computer Science Capstone Project CS 691

Prof. Kshitij Sharma

Spring 2026

Group 8:

Sharvani Kadarla

Deepali Sawant

Jai Sheel Bhasker Paladugu

Ashish Reddy Majjigapu

Agenda

1. Introduction

- ☐ Team Member Roles & Responsibilities

2. Improvements from Professor Feedback

- ☐ Updates made based on feedback

3. Project Overview

- ☐ Project Description
- ☐ Personas
- ☐ Minimum Viable Product (MVP)

4. System Design & Technology

- ☐ Technologies Used
- ☐ Algorithms
- ☐ System Diagrams

5. Team Collaboration

- ☐ Team Working Agreement

6. Product Development

- ☐ Product Backlog - Acceptance Criteria & User Stories
- ☐ Sprint 1 Backlog

7. Sprint Progress & Metrics

- ☐ Test Cases
- ☐ Completed & Pending Stories
- ☐ Team Velocity & Burndown Chart

8. Sprint Review

- ☐ Sprint Retrospective
- ☐ Sprint 2 Planning

9. Project Demonstration

- ☐ Application Screenshots
- ☐ API Demonstration

10. GitHub Repository & Live Application Demo

Team Member Roles and Responsibilities

Our project success hinges on a clearly defined structure, where each team member's expertise drives specific outcomes.

Sharvani Kadarla **Project Manager & Software Developer**

Define project vision, roadmap, and milestones

Plan sprints and coordinate team tasks

Design system architecture and data flow

Ensure integration across frontend, backend, and AI components

Monitor project progress and resolve blockers

Deepali Sawant **Frontend & Visualization Engineer**

Develop interactive UI and dashboards

Build portfolio and knowledge graph visualizations

Ensure clean, user-friendly UI/UX design

Integrate frontend with backend APIs and AI outputs

Jai Sheel Bhasker Paladugu **AI / Machine Learning Engineer**

Develop AI models for event impact analysis

Build causal graphs for industry and company effects

Implement LLM-based explainable insights

Run scenario simulations for portfolio risk

Ashish Reddy Majjigapu **Backend & Data Engineering Lead**

Develop RESTful APIs using FastAPI

Build data ingestion and preprocessing pipelines

Manage data storage and validation

Support portfolio risk calculations and backend performance

Improvements made from Professor's Feedback

Post the Sprint 0 presentation, we received feedback from the professor regarding team communication and meeting coordination. Based on the feedback, we improved our communication process and established a consistent weekly meeting schedule for better collaboration.

We also organized tasks more clearly during sprint planning and improved progress tracking and responsibility sharing within the team. These changes helped the team stay aligned and work more efficiently on the RippleWealth project.

Problem Statement

- ❑ Modern investment platforms are reactive — they alert investors only after stock prices change
- ❑ They do not consider early real-world signals like supply chain disruptions, weather events, labor strikes, or logistics delays
- ❑ Current tools focus mainly on financial indicators and ignore non-financial risk triggers
- ❑ Investors lack clear explanations of why their portfolio risk is increasing
- ❑ As a result, users cannot anticipate risk or make proactive portfolio decisions

Problem Description

RippleWealth is an AI-driven investment risk intelligence platform that anticipates market risks by detecting real-world disruptions—such as weather events, supply chain delays, and labor strikes—before they appear in stock prices. Unlike traditional investment tools that react only after price changes, RippleWealth provides early insights to help investors understand potential portfolio impacts.

Using a multi-agent AI architecture and LLM-based causal reasoning, the platform maps how these events propagate across industries and impact companies and portfolios. Through an interactive Knowledge Graph and explainable risk metrics like Value-at-Risk (VaR), RippleWealth enables proactive, transparent, and data-driven portfolio risk management.

Team Working Agreement

Participation

- ❑ Attend scheduled meetings on time
- ❑ Come prepared with progress updates
- ❑ Inform the team leader in advance if absent

Meetings

- ❑ Weekly meetings via Zoom / Google Meet
- ❑ Team leader schedules and shares meeting links
- ❑ Meeting minutes recorded after each session

Work Quality & Collaboration

- ❑ Share work early for feedback
- ❑ Maintain professionalism and accountability
- ❑ Ensure alignment with project goals

Work Division & Ownership

- ❑ Tasks assigned during sprint planning
- ❑ Equal responsibility among team members
- ❑ Meet agreed deadlines and communicate blockers early

Communication Tools

- ❑ Zoom / Google Meet – Sprint meetings
- ❑ WhatsApp – Quick updates and urgent communication
- ❑ Google Docs & Slides – Collaborative deliverables

Team Transparency

- ❑ Cameras encouraged during meetings
- ❑ Maintain open communication and team trust

User Persona 1: Ethan Miller – Retail Investor (Young Professional)

Ethan represents our younger target user: a data-driven professional eager for sophisticated, simple tools to manage his growing portfolio.



DEMOGRAPHICS

Name: Ethan Miller
Age: 29
Location: Austin, Texas
Job: Software Engineer
Salary: \$95,000–\$120,000/year
Family: Lives with partner
Investment Level: Beginner to Intermediate Investor

GOALS

- ❑ Understand Value-at-Risk clearly
- ❑ Simulate market crash scenarios
- ❑ Get simple portfolio risk scores
- ❑ Improve long-term returns
- ❑ Make safer investment decisions

FRUSTRATIONS

- ❑ Cannot clearly measure portfolio risk
- ❑ Hard to understand downside scenarios
- ❑ Too many scattered finance tools
- ❑ Confusing probability models
- ❑ Lacks simple visual risk insights

INTERESTS

- ❑ Personal finance and investing
- ❑ Portfolio tracking apps
- ❑ Market trend analysis
- ❑ Tech and automation tools
- ❑ Financial podcasts
- ❑ Long-term wealth building

User Persona 2: Sophia Chen – Financial Risk Analyst (Professional User)

Sophia represents our advanced professional user: an expert operating in a high-stakes, data-intensive environment where deep analytics and systemic accuracy are critical.



DEMOGRAPHICS

Name: Sophia Chen
Age: 38
Location: New York, NY
Job: Financial Risk Analyst
Salary: \$130,000–\$160,000/year
Family: Married, one child
Investment Level: Advanced / Professional

GOALS

- ❑ Run large-scale Monte Carlo simulations
- ❑ Identify systemic risk pathways
- ❑ Model inter-company relationships
- ❑ Generate client risk reports quickly
- ❑ Improve predictive accuracy

FRUSTRATIONS

- ❑ Manual scenario modeling is time-consuming
- ❑ Tools lack integrated causal analysis
- ❑ Hard to map indirect market impacts
- ❑ Needs faster simulation turnaround
- ❑ Limited explainability in some models

INTERESTS

- ❑ Risk modeling techniques
- ❑ Market stress testing
- ❑ Portfolio optimization
- ❑ Quantitative finance
- ❑ Economic indicators
- ❑ Financial data visualization

User Persona 3: Carlos Ramirez – Business Owner / Wealth Planner

Carlos represents our entrepreneur user: a strategic investor who manages complex business and personal holdings and needs to see the "cause-and-effect" of global economic events on his investments.



DEMOGRAPHICS

Name: Carlos Ramirez

Age: 45

Location: Atlanta, Georgia

Job: Small Business Owner

Salary: \$110,000–\$150,000/year

Family: Married

Investment Level: Moderate Investor

GOALS

- ❑ Understand event → market → portfolio impact
- ❑ Visualize causal financial relationships
- ❑ Stress-test business investments
- ❑ Reduce exposure to high-risk sectors
- ❑ Make resilient investment plans

FRUSTRATIONS

- ❑ Unsure how world events affect investments
- ❑ No clear impact chain visibility
- ❑ Market shocks feel unpredictable
- ❑ Hard to evaluate sector dependencies
- ❑ Lacks scenario-based planning tools

INTERESTS

- ❑ Business growth strategy
- ❑ Market and sector trends
- ❑ Supply chain news
- ❑ Financial planning tools
- ❑ Diversified investments
- ❑ Economic forecasts

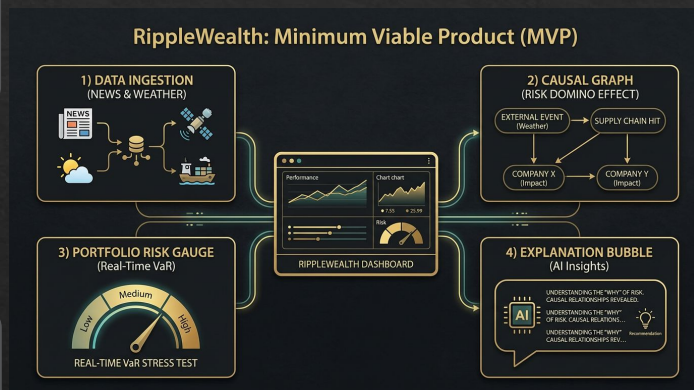
Minimal Viable Product (MVP)

Alternative Data Engine

"Monitors news, weather, supply chain, and labor reports for early 'Butterfly Events' (strikes, delays, floods) using simple keyword matching and public APIs"

Causal Chain Prototype

"Visual causal graph showing Event → Sector → Company impact paths so users see exactly how disruptions ripple through their portfolio holdings"



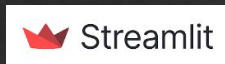
Portfolio Risk Dashboard

"Real-time Value-at-Risk (VaR) and risk scores for portfolios using basic Monte Carlo simulation on mock market data and detected events"

Explainable Insights

"Plain English explanations connecting events to risk changes: 'Port strike → Auto sector delay → Your GM holding risk ↑12%'"

Core Technologies Powering Our Platform



Backend & APIs

Python + FastAPI

- REST APIs for portfolios & risk data
- User authentication
- Frontend connection

Data & Storage

SQLite Database

- User portfolios & events storage
- CSV/mock API ingestion
- Data preprocessing

AI & Analytics

Python Analytics

- Monte Carlo simulation (VaR)
- Keyword NLP for events
- Causal graph prototype

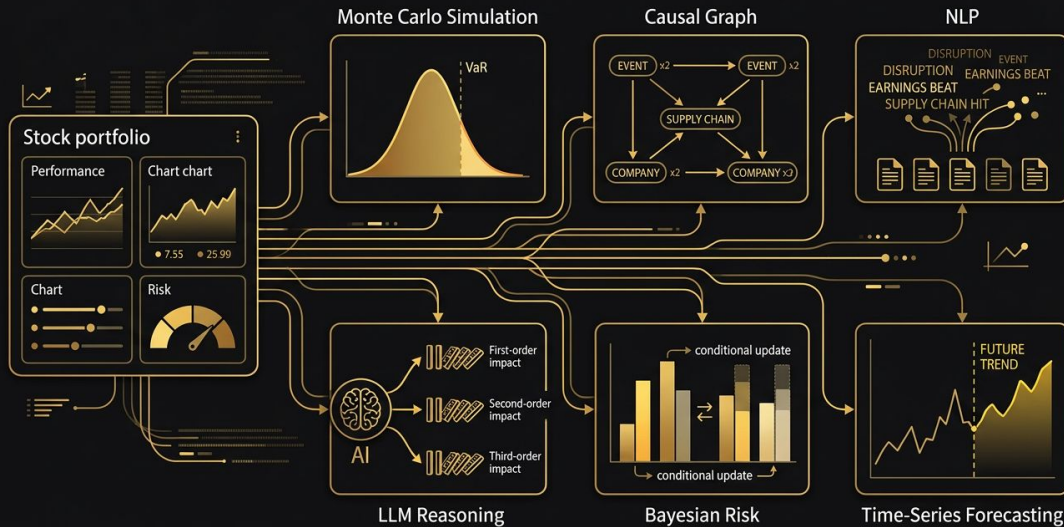
Frontend & Visualization

Streamlit Dashboard

- Interactive risk views
- Charts for events & paths
- Responsive UI testing

Algorithms

AI & QUANTITATIVE FINANCE TECHNIQUES FOR INVESTMENT RISK PLATFORM



Monte Carlo Simulation — Portfolio Risk Modeling

- Computes portfolio Value-at-Risk (VaR)
- Simulates thousands of possible market stress scenarios
- Estimates potential downside loss under uncertainty

Causal Graph Modeling (Graph Theory)

- Builds event → supply chain → company relationship graphs
- Identifies direct and indirect impact paths
- Enables disruption impact tracing across industries

Natural Language Processing (NLP)

- Extracts “Butterfly Events” from alternative data sources
- Processes news, weather, and labor reports
- Converts unstructured text into structured risk signals

Large Language Model (LLM) Reasoning

- Simulates second- and third-order domino effects
- Performs causal chain reasoning across sectors
- Generates explainable impact narratives

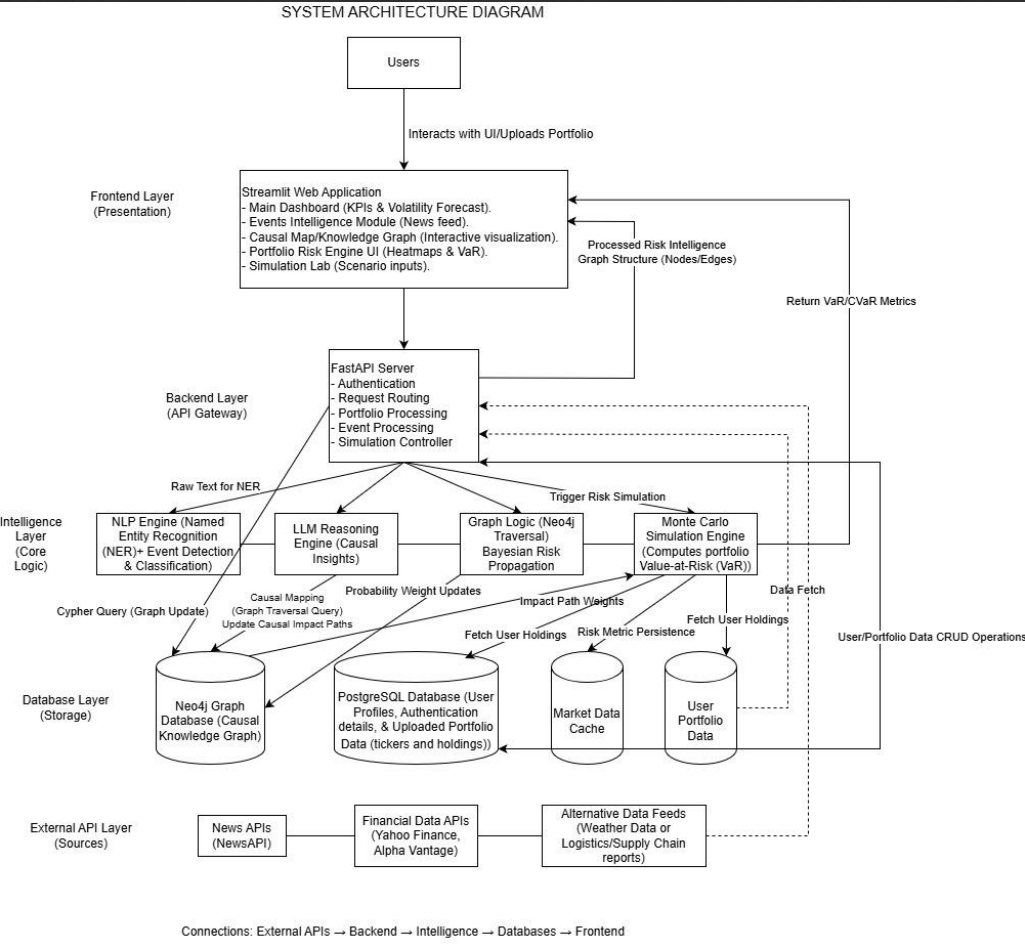
Bayesian Risk Propagation

- Assigns probability weights to impact chains
- Updates risk exposure dynamically as new signals arrive
- Supports uncertainty-aware decision modeling

Time-Series Forecasting (ARIMA / LSTM)

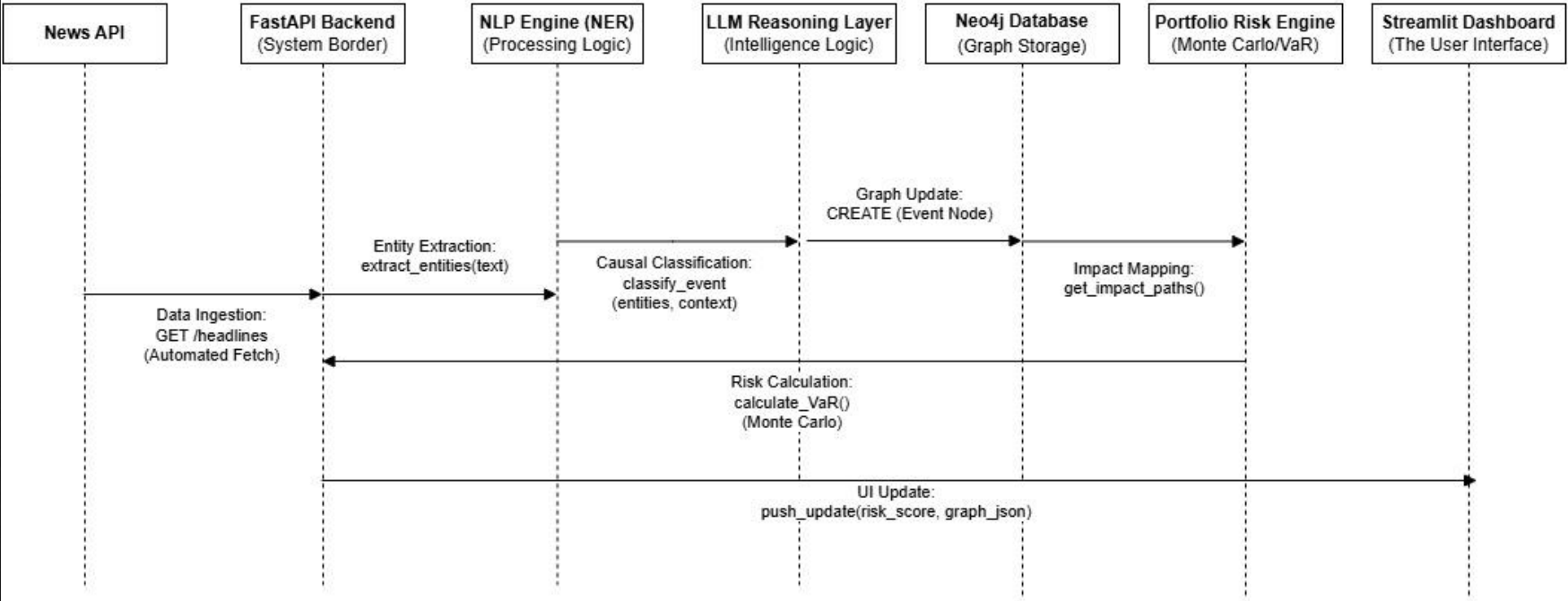
- Predicts short-term volatility trends
- Supports stress scenario projections
- Enhances forward risk estimation

System Architecture Diagram

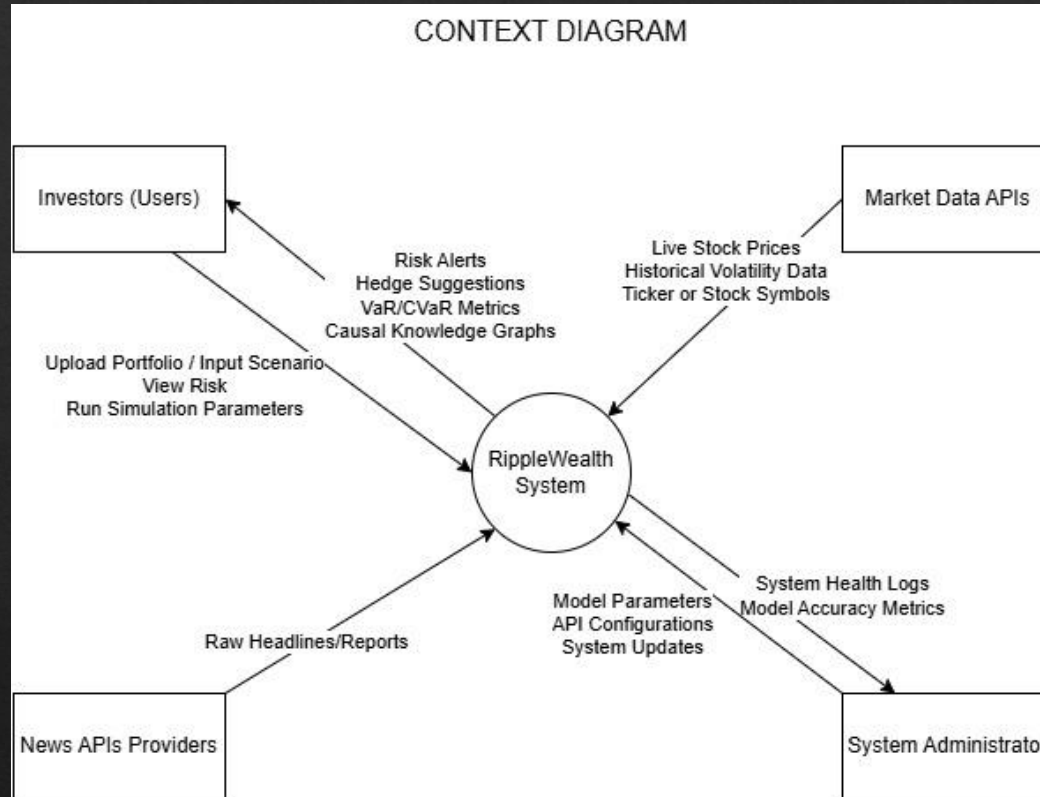


Sequence Diagram

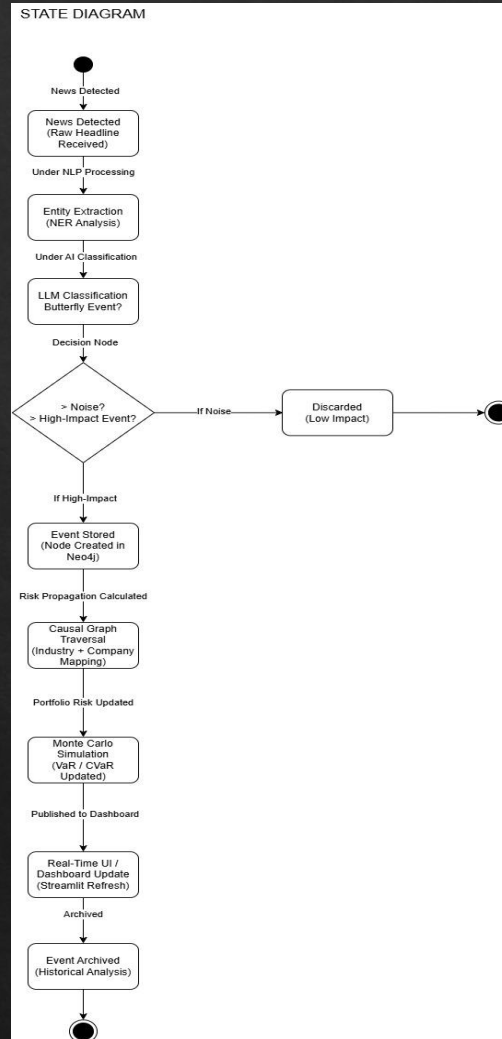
SEQUENCE DIAGRAM



Context Diagram



State Diagram



Product Backlog - User Stories

Spaces

RippleWealth

...

Summary

Timeline

Backlog

Board

Calendar

List

Forms

Development

Code

Archived work items

Pages

More 2

+

Search backlog

DS

AM

JP

SK

Filter

Team-8 Sprint 1

24 Feb – 10 Mar (21 work items)

51 10 27 Complete sprint ...

Front-end Design, Create MVP Prototype, Create System Diagrams, Create Acceptance Criteria, Create Test Cases, Create Mock-up Web Design Page, Create User Stories or Tasks, Work on Deliverable 2, Working API, Application ...

T8-41	As a potential investor, I want to understand RippleWealth's proactive risk intelligence approach to help me decide w...	TO DO	2	
T8-44	As an investor, I want to view a projected portfolio value chart after simulation so that I can visually understand the ...	TO DO	3	
T8-47	As an investor, I want to see portfolio risk metrics so that I can understand potential exposure to market volatility.	TO DO	3	
T8-50	As a user, I want to view event details such as location, industry impact, and severity so that I can better understand...	TO DO	8	
T8-51	As an investor, I want to run an event simulation so that I can understand the potential impact of a global event on m...	TO DO	3	
T8-52	As an investor, I want to compare my current portfolio with recommended adjustments so that I can rebalance my in...	TO DO	3	
T8-62	As an investor, I want to receive investment suggestions based on global events so that I can reduce portfolio risk.	TO DO	8	
T8-63	As an investor, I want to understand Value at Risk so that I can estimate the maximum likely loss in normal market co...	TO DO	8	
T8-64	As a portfolio manager, I want to receive automated rebalancing suggestions, so that I can reduce my exposure to hi...	TO DO	5	
T8-65	As a user, I want to browse and filter real-world disruption events, so that I can stay informed about specific crises ...	TO DO	3	
T8-66	As an investor, I want early investment advice so that I can move my money to safer assets when the market becom...	TO DO	5	

Acceptance Criteria

Issue Type	ID	Summary	Scenario
Acceptance Criteria	T8-41	As a potential investor, I want to understand RippleWealth's proactive risk intelligence approach to help me decide whether the platform meets my portfolio management needs.	Given I am a visitor on the RippleWealth landing page When the landing page loads Then the system displays the platform headline and description of proactive risk intelligence And the system displays feature sections explaining disruption detection, causal modeling, and analytics
Acceptance Criteria	T8-44	As an investor, I want to view a projected portfolio value chart after simulation so that I can visually understand the effect of the selected event.	Given I have run an event simulation When the simulation results are generated Then the system displays a projected portfolio value chart And the chart reflects the impact of the selected event
Acceptance Criteria	T8-47	As an investor, I want to see portfolio risk metrics so that I can understand potential exposure to market volatility	Given I am on the Risk Metrics page When the page loads Then the system displays portfolio risk metrics such as Value at Risk (VaR), Conditional Value at Risk (CVaR), and sector exposure Given the risk metrics are displayed When I review the metrics Then the system clearly labels each metric And the system provides a short description explaining what each metric represents
Acceptance Criteria	T8-50	As a user, I want to view event details such as location, industry impact, and severity so that I can better understand the disruption.	Given I am on the Events page When the list of disruption events is displayed Then the system shows basic information for each event Given I select or open an event When the event details page loads Then the system displays the event location, affected industry, and severity level And the system displays additional information that helps explain the disruption
Acceptance Criteria	T8-51	As an investor, I want to run an event simulation so that I can understand the potential impact of a global event on my portfolio value.	Given I am on the Event Simulation page When I select an event and click "Run Simulation" Then the system simulates the impact of the selected event on the portfolio And the system displays a projected portfolio value chart

Acceptance Criteria

Issue Type	ID	Summary	Scenario
Acceptance Criteria	T8-52	As an investor, I want to compare my current portfolio with recommended adjustments so that I can rebalance my investments to reduce risk.	Given I am viewing investment suggestions When the system generates recommendations Then the system displays my current portfolio allocation And the system displays a recommended portfolio allocation And the system provides a message explaining the suggested adjustment strategy
Acceptance Criteria	T8-62	As an investor, I want to receive investment suggestions based on global events so that I can reduce portfolio risk.	Given I am on the Investment Suggestions page When I select an event and click "Get Suggestions" Then the system analyzes the event impact on industries And the system displays assets to avoid And the system displays recommended investments
Acceptance Criteria	T8-63	As an investor, I want to understand Value at Risk so that I can estimate the maximum likely loss in normal market conditions.	Given I am on the Risk Metrics page When the metrics section is displayed Then the system shows the Value at Risk value And the system provides a label or description explaining the meaning of the metric
Acceptance Criteria	T8-64	As a portfolio manager, I want to receive automated rebalancing suggestions, so that I can reduce my exposure to high-risk sectors identified by the AI.	Given I am a signed-in portfolio manager And my portfolio has been analyzed by the system When the system detects high-risk sector exposure Then the system generates automated rebalancing suggestions And the system displays recommended assets to reduce exposure And the system suggests alternative investments to rebalance the portfolio
Acceptance Criteria	T8-65	As a user, I want to browse and filter real-world disruption events, so that I can stay informed about specific crises affecting my investments.	Given I am on the Global Events page When the list of disruption events loads Then the system displays a list of real-world disruption events And each event shows details such as industry, severity level, and confidence score
Acceptance Criteria	T8-66	As an investor, I want early investment advice so that I can move my money to safer assets when the market becomes unstable.	Given I am on the Investment Suggestions page When a disruption event is selected Then the system analyzes the impact of the event on industries and assets Given the analysis is completed When investment suggestions are generated Then the system displays recommended safer assets And the system displays assets that may be risky during the disruption And the system provides a suggested portfolio adjustment strategy

Test Cases

Test Case ID	Module	Test Objective	Test Steps	Expected Result
TC01	Homepage	Verify the Home dashboard loads correctly	1. Launch the RippleWealth application 2. Navigate to Home from sidebar	Home page loads displaying platform description and workflow steps
TC02	Navigation	Verify sidebar navigation works	1. Click Portfolio in sidebar 2. Click Events 3. Click Risk Metrics	Each screen opens correctly when selected
TC03	Portfolio	Verify portfolio table is displayed	1. Navigate to Portfolio page 2. View the sample portfolio table	Table displays assets, ticker symbols, and investment amounts
TC04	Portfolio	Verify total portfolio value is displayed	1. Open Portfolio screen 2. Check value shown above table	Correct total portfolio value is displayed
TC05	Events	Verify global events list is displayed	1. Navigate to Events screen	List of global events with industry, severity, and confidence appears
TC06	Risk Metrics	Verify risk metrics are calculated	1. Navigate to Risk Metrics page	VaR, Conditional VaR, and sector exposure values are displayed
TC07	Risk Metrics	Verify metric legend is shown	1. Scroll within Risk Metrics page	Explanation of VaR, CVaR, and sector exposure is visible
TC08	Simulation	Verify event selection dropdown	1. Navigate to Simulation page 2. Click event dropdown	List of available events appears
TC09	Simulation	Verify simulation execution	1. Select an event 2. Click Run Simulation	Projected portfolio value chart is generated
TC10	Simulation	Verify projected loss display	1. Run a simulation 2. Scroll to results section	Estimated loss (\$) and estimated loss (%) are displayed
TC11	Suggestions	Verify investment suggestions	1. Navigate to Suggestions page 2. Select event 3. Click Get Suggestions	Suggested investments and assets to avoid are displayed
TC12	Suggestions	Verify recommendation explanation	1. Run suggestion analysis	Recommendation note explaining diversification strategy appears



Sprint Backlog: Sprint 2

☐ Team-8 Sprint 2 11 Mar – 7 Apr (11 work items)	000	Start sprint	...
T8-20 Setup FastAPI Project & Database Models	TO DO ▾	-	SK
T8-21 Implement Portfolio & Risk API Endpoints	TO DO ▾	-	JP
T8-22 Causal Event Modeling	TO DO ▾	-	JP
T8-23 Knowledge Graph Visualization Structure	TO DO ▾	-	DS
T8-24 LLM Reasoning Logic	TO DO ▾	-	AM
T8-25 Event Impact Simulation	TO DO ▾	-	AM
T8-26 AI-Backend Integration	TO DO ▾	-	JP
T8-27 Draft Technical Paper	TO DO ▾	-	SK
T8-28 Develop Frontend	TO DO ▾	-	DS
T8-29 Develop Backend	TO DO ▾	-	AM
T8-30 Work on Deliverable 3	TO DO ▾	-	SK

Sprint Backlog: Sprint 3

☐ Team-8 Sprint 3 8 Apr – 5 May (7 work items)	000	Start sprint	...
☐  T8-31 Finish Development of Frontend	TO DO ▾	-	
 T8-32 Finish Development of Backend	TO DO ▾	-	
 T8-33 Frontend-Backend Integration Working	TO DO ▾	-	
 T8-34 System Testing	TO DO ▾	-	
 T8-35 Finish Project Demo	TO DO ▾	-	
 T8-36 Finish Technical Paper	TO DO ▾	-	
 T8-37 Work on Deliverable 4	TO DO ▾	-	

Stories Completed - Sprint 1

☐ Team-8 Sprint 1 24 Feb - 10 Mar (21 work items)

0 10 78 Complete sprint ...

Front-end Design, Create MVP Prototype, Create System Diagrams, Create Acceptance Criteria, Create Test Cases, Create Mock-up Web Design Page, Create User Stories or Tasks, Work on Deliverable 2, Working API, Application ...

 T8-41	As a potential investor, I want to understand RippleWealth's proactive risk intelligence approach to help me decide w...	DONE ✓	2	
 T8-44	As an investor, I want to view a projected portfolio value chart after simulation so that I can visually understand the ...	DONE ✓	3	
 T8-47	As an investor, I want to see portfolio risk metrics so that I can understand potential exposure to market volatility.	DONE ✓	3	
 T8-50	As a user, I want to view event details such as location, industry impact, and severity so that I can better understand...	DONE ✓	8	
 T8-51	As an investor, I want to run an event simulation so that I can understand the potential impact of a global event on m...	DONE ✓	3	
 T8-52	As an investor, I want to compare my current portfolio with recommended adjustments so that I can rebalance my in...	DONE ✓	3	
 T8-62	As an investor, I want to receive investment suggestions based on global events so that I can reduce portfolio risk.	DONE ✓	8	
 T8-63	As an investor, I want to understand Value at Risk so that I can estimate the maximum likely loss in normal market co...	DONE ✓	8	
 T8-64	As a portfolio manager, I want to receive automated rebalancing suggestions, so that I can reduce my exposure to hi...	DONE ✓	5	
 T8-65	As a user, I want to browse and filter real-world disruption events, so that I can stay informed about specific crises ...	DONE ✓	3	
 T8-66	As an investor, I want early investment advice so that I can move my money to safer assets when the market becom...	DONE ✓	5	

Sprint 1 Schedule

☐ Team-8 Sprint 1 24 Feb – 10 Mar (23 work items)	01577	Complete sprint	...
Front-end Design, Create MVP Prototype, Create System Diagrams, Create Acceptance Criteria, Create Test Cases, Create Mock-up Web Design Page, Create User Stories or Tasks, Work on Deliverable 2, Working API, Applicati...			
 T8-10 Front-end Design	IN PROGRESS ▾	5	DS
 T8-11 Create MVP Prototype	DONE ▾	8	AM
 T8-12 Create System Diagrams	DONE ▾	5	SK
 T8-13 Create Acceptance Criteria	DONE ▾	2	DS
 T8-14 Create Test Cases	DONE ▾	2	DS
 T8-15 Create Mock-up Web Design Page	DONE ▾	3	JP
 T8-16 Create User Stories or Tasks	DONE ▾	2	AM
 T8-17 Work on Deliverable 2	DONE ▾	2	SK
 T8-18 Working API	IN PROGRESS ▾	5	SK
 T8-19 Application Demo	DONE ▾	3	JP

Metrics - Status Overview as of Sprint 1

Spaces



RippleWealth



Summary



Timeline



Backlog



Board



Calendar



List



Forms



Development



Code

More 4

Status overview

Get a snapshot of the status of your work items. [View all work items](#)



Done: 29

In Progress: 3

To Do: 29

Metrics - Sprint 1 Burndown Chart

Spaces / RippleWealth / Reports

Sprint burndown chart

> How to read this report

Sprint

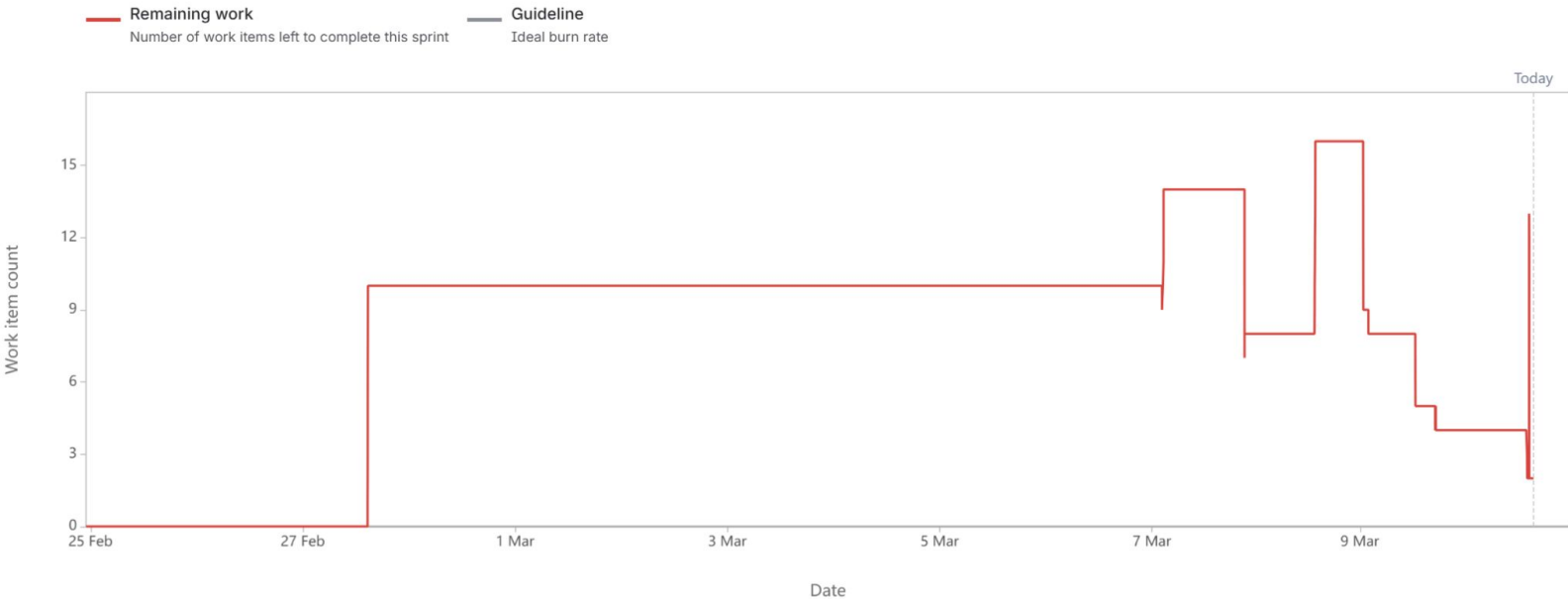
Team-8 Sprint 1

Estimation field

Work item count

Date - 24 February 2026 - 10 March 2026

Sprint goal - Front-end Design, Create MVP Prototype, Create System Diagrams, Create Acceptance Criteria, Create Test Cases, Create Mock-up Web Design Page, Create User Stories or Tasks, Work on Deliverable 2, Working API, Application Demo



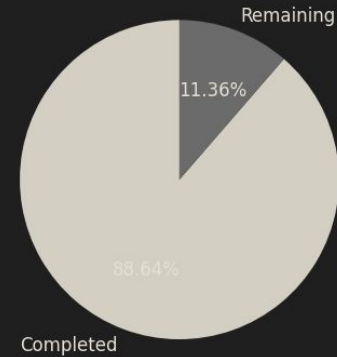
Metrics - Sprint 1 Ratio Status

Committed Stories Points: 88

Completed Stories Points: 78

Completed/Committed Ratio (Story Points) = $78/88$,
which is 88.63%

Sprint 1 Story Points Completion (88.63%)



Retrospective



What Went Well

By adopting Slack as our primary operational hub, the team now receives instantaneous notifications for every update made in Jira and GitHub. This integration has successfully bridged the gap between our AI logic, backend services, and frontend visualization, ensuring everyone stays aligned with the RippleWealth vision. The team showed strong collaboration and adaptability in addressing initial technical challenges.



Action Items for Next Sprint

To improve transparency, every weekly meeting will now require each member to present a concise summary of their technical accomplishments. Furthermore, we are instituting a mandatory mid-sprint progress check-in to proactively identify and resolve any technical blockers before they impact our final deliverables.



What Could Be Improved

We have identified that the submission of specific deliverables must strictly adhere to our internal deadlines. During this sprint, several late submissions occurred, which created minor bottlenecks in our overall development velocity.



Sprint 2 Schedule

☐ Team-8 Sprint 2 11 Mar – 7 Apr (11 work items)	000	Start sprint	...
T8-20 Setup FastAPI Project & Database Models	TO DO ▾	-	SK
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T8-28 Develop Frontend	TO DO ▾	-	DS
T8-29 Develop Backend	TO DO ▾	-	AM
T8-30 Work on Deliverable 3	TO DO ▾	-	SK

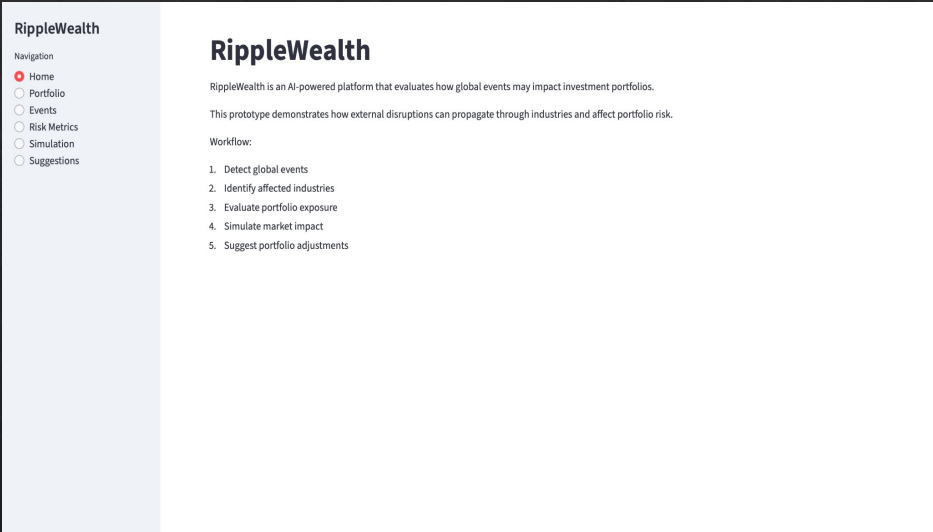
Stories planned and committed for Sprint 2

☐ Team-8 Sprint 2 11 Mar – 7 Apr (23 work items)	66 2 0	Start sprint	...
T8-48 As an investor, I want to see my portfolio's Value-at-Risk (VaR), so that I understand the potential losses I could face...	TO DO ▼	8	
T8-53 As a user, I want the system to update risk predictions when new events occur, so that my portfolio risk assessment ...	TO DO ▼	5	
T8-54 As an investor, I want the system to calculate potential portfolio losses using Monte Carlo simulations, so that I can ...	TO DO ▼	8	
T8-55 As an investor, I want the platform to generate explanations for its risk predictions, so that I can understand the reas...	TO DO ▼	8	
T8-56 As an investor, I want to upload my portfolio holdings, so that the system can analyze how global events may affect ...	TO DO ▼	3	
T8-57 As an investor, I want the system to calculate potential portfolio losses using Monte Carlo simulations, so that I can ...	TO DO ▼	5	
T8-58 As an investor, I want to simulate hypothetical events like natural disasters or geopolitical crises, so that I can under...	TO DO ▼	8	
T8-59 As a developer, I want to implement Portfolio & Risk API Endpoints, so that the frontend can securely request and re...	TO DO ▼	5	
T8-60 As an investor, I want the platform to generate natural language explanations for its risk predictions, so that I can un...	TO DO ▼	8	
T8-61 As a user, I want to view a centralized risk dashboard with a sector exposure heatmap, so that I can visually identify...	TO DO ▼	5	
T8-45 As a portfolio manager, I want to filter events by region, severity or industry so that I can focus only on disruptions r...	TO DO ▼	3	
T8-43 As as registered user, I want to recover my password, so that I can regain access to my account without losing data.	IN PROGRESS ▼	2	

Project Demo – Sprint 1 (Current Sprint)



Screenshots of The Working Prototype



"Our system uses FastAPI to expose REST APIs for portfolio data, risk analytics, simulation, and AI-based investment recommendations. The frontend dashboard interacts with these APIs to dynamically update risk scores and portfolio projections."

RippleWealth API

0.1.0

OAS 3.1

/openapi.json

default

GET

/events Get Events

GET

/portfolio Portfolio

GET

/portfolio/risk Portfolio Risk

POST

/simulate Simulate

POST

/recommend Recommend

POST

/rebalance Rebalance

Schemas

HTTPValidationError >

Expand all

object

ValidationError >

Expand all

object



Live Application Demo

GitHub Wiki Page Link



[SharvaniKadarla/RippleWealth Wiki Page \(github.com\)](https://github.com/SharvaniKadarla/RippleWealth/wiki)

View on GitHub

THANK YOU